Stewart, Calculus: Early Transcendentals

Section 15.9, Exercise 45: the volume of a bumpy sphere

Problem

The surfaces $\rho = 1 + \frac{1}{5}\sin m\theta \, \sin n\varphi$ have been used as models for tumors. The "bumpy sphere" with $m = 6$ and $n = 5$ is shown. Use a computer algebra system to find the volume it encloses.

Symbols used in this solution

- ρ, θ, φ are spherical coordinates: ρ is the distance from the origin, φ is the angle from the positive z-axis and θ is the angle in the xy-plane, measured from the positive x-axis. A point has $x = \rho\sin\varphi\cos\theta$, $y = \rho\sin\varphi\sin\theta$, $z = \rho\cos\varphi$.
- The surface is written $\rho = \rho_{\max}(\theta, \varphi)$, where $\rho_{\max} = 1 + \frac{1}{5}\sin 6\theta \, \sin 5\varphi$.
- E is the solid that the surface encloses, and dV is the volume element. In spherical coordinates $dV = \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta$. The factor $\rho^2 \sin\varphi$ is the Jacobian, and it is the reason a spherical volume integral is not simply $\iiint d\rho \, d\varphi \, d\theta$.
- V is the volume of E.

Part (a): the exact volume

Step 1. The region and the order of integration

Because $|\sin 6\theta \, \sin 5\varphi| \le 1$, the radius satisfies

$$0.8 \le \rho_{\max}(\theta, \varphi) \le 1.2. \tag{1}$$

The surface is therefore star shaped about the origin: each ray that leaves the origin in a direction (θ, φ) meets the surface exactly once, at distance $\rho_{\max}(\theta, \varphi)$. Hence the solid is described by

$$E = \Big\{(\rho, \theta, \varphi) : 0 \le \rho \le \rho_{\max}(\theta, \varphi), \ 0 \le \varphi \le \pi, \ 0 \le \theta \le 2\pi\Big\}, \tag{2}$$

and this description is legal because every ray gives one interval of ρ. The angular ranges are the full ranges of spherical coordinates: a half turn from the north pole to the south pole in φ, and a full turn in θ. The order $d\rho \, d\varphi \, d\theta$ is used, so the inner integral runs along a ray.

Step 2. Integrate in ρ first

The ρ integral is elementary, because the upper limit enters only as a power:

$$\int_0^{\rho_{\max}} \rho^2 \sin\varphi \, d\rho = \frac{\rho_{\max}^3}{3} \sin\varphi. \tag{3}$$

Therefore

$$V = \int_0^{2\pi} \int_0^{\pi} \frac{\rho_{\max}^3}{3} \sin\varphi \, d\varphi \, d\theta = \frac{1}{3} \iint_E \rho^3 \sin\varphi \, d\varphi \, d\theta. \tag{4}$$

Equation (4) is the general formula for the volume enclosed by a surface given in the form $\rho = \rho(\theta, \varphi)$. The volume is $\frac{4}{3}\pi$ times the average value of ρ^3 over the sphere of directions.

Step 3. Expand the cube

Write $u = \frac{1}{5} \sin 6\theta \, \sin 5\varphi$, so that $\rho_{\max} = 1 + u$ and

$$\rho_{\max}^3 = 1 + 3u + 3u^2 + u^3 = 1 + \frac{3}{5} \sin 6\theta \sin 5\varphi + \frac{3}{25} \sin^2 6\theta \, \sin^2 5\varphi + \frac{1}{125} \sin^3 6\theta \, \sin^3 5\varphi. \tag{5}$$

Each term is a product of a function of θ and a function of φ, so each term of the double integral separates into a product of two one dimensional integrals.

Step 4. The four terms, one at a time

(i) The constant term.

$$\int_0^{2\pi} \int_0^{\pi} \sin\varphi \, d\varphi \, d\theta = 2\pi \cdot \Big[-\cos\varphi \Big]_0^{\pi} = 4\pi. \tag{6}$$

(ii) The linear term vanishes. The factor $\sin 6\theta$ completes six full periods on $[0, 2\pi]$, so its integral is zero:

$$\int_0^{2\pi} \sin 6\theta \, d\theta = 0 \quad \Longrightarrow \quad \frac{3}{5} \int_0^{2\pi} \int_0^{\pi} \sin 6\theta \sin 5\varphi \sin\varphi \, d\varphi \, d\theta = 0. \tag{7}$$

The reason is symmetry, not cancellation of round-off: over one full period the positive and the negative half of a sine wave have equal area.

(iii) The square term. Here both factors are non-negative, so nothing vanishes:

$$\int_0^{2\pi} \sin^2 6\theta \, d\theta = \pi, \qquad \int_0^{\pi} \sin^2 5\varphi \, \sin\varphi \, d\varphi = \frac{100}{99}. \tag{8}$$

The first value is the standard half-period average $\frac{1}{2} \cdot 2\pi$. The second one is computed with $\sin^2 5\varphi = \frac{1}{2}(1 - \cos 10\varphi)$:

$$\int_0^{\pi} \sin^2 5\varphi \sin\varphi \, d\varphi = \frac{1}{2} \int_0^{\pi} \sin\varphi \, d\varphi - \frac{1}{2} \int_0^{\pi} \cos 10\varphi \sin\varphi \, d\varphi = 1 - \frac{1}{2}\Big(-\frac{2}{99}\Big) = 1 + \frac{1}{99}, \tag{9}$$

because the product-to-sum formula gives $\cos 10\varphi \sin \varphi = \frac{1}{2}(\sin 11\varphi - \sin 9\varphi)$ and

$$\int_0^{\pi} \cos 10\varphi \sin \varphi \, d\varphi = \frac{1}{2}\Big(\frac{2}{11} - \frac{2}{9}\Big) = -\frac{2}{99}. \tag{10}$$

The square term therefore contributes

$$\frac{3}{25} \cdot \pi \cdot \frac{100}{99} = \frac{4\pi}{33}. \tag{11}$$

(iv) The cubic term vanishes. It contains the factor $\sin^3 6\theta$, and

$$\int_0^{2\pi} \sin^3 6\theta \, d\theta = \int_0^{2\pi} \big(1 - \cos^2 6\theta\big) \sin 6\theta \, d\theta = \Big[-\frac{\cos 6\theta}{6} + \frac{\cos^3 6\theta}{18}\Big]_0^{2\pi} = 0, \tag{12}$$

because both antiderivatives have the same value at 0 and at 2π. The whole cubic term is zero, whatever the φ integral is.

Step 5. The volume

Add the four contributions and divide by 3:

$$\boxed{V = \frac{1}{3}\Big(4\pi + 0 + \frac{4\pi}{33} + 0\Big) = \frac{4\pi}{3} \cdot \frac{34}{33} = \frac{136\pi}{99} \approx 4.3157} \tag{13}$$

As a check on the size: the unit sphere has volume $\frac{4}{3}\pi \approx 4.1888$, and $\frac{136\pi}{99}$ is larger by the factor $\frac{34}{33} \approx 1.0303$. That is expected. The average of ρ over all directions is exactly 1 (the linear term vanishes), but ρ^3 is a convex function of ρ, so its average is larger than 1^3.

Step 6. Checks

Table 1 lists the checks. The exact value was obtained twice with `sympy`: once term by term as above, and once from the direct triple integral. The numeric values are independent: a two dimensional Gauss–Legendre rule with the ρ integral done exactly, a full three dimensional Gauss–Legendre rule in (ρ, φ, θ), and a Monte Carlo estimate inside the ball of radius 1.2.

Table 1: Checks of $V = \frac{136\pi}{99}$ for the bumpy sphere with $m = 6$, $n = 5$.

Method	Value	Error
exact, `sympy` (term by term)	$136\pi/99 = 4.3157232413$	0
exact, `sympy` (direct triple integral)	4.3157232413	0
2D Gauss–Legendre, exact in ρ	4.315723241295	1.7×10^{-14}
3D Gauss–Legendre	4.315723241295	2.7×10^{-15}
Monte Carlo, 4×10^6 points	4.3168 ± 0.0018	1.1×10^{-3}
same code with $\rho_{\max} = 1$ (control)	$4\pi/3 = 4.1887902048$	$< 10^{-10}$

Step 7. The picture

Figure 1 is the surface itself, drawn with the same style as the textbook picture: a closed surface, coloured by the distance from the origin, with no axes. Six bumps run around the equator and five rows of bumps run from pole to pole, because $\sin 6\theta$ completes six periods in a full turn and $\sin 5\varphi$ completes five half periods between the poles. Figure 2 shows where the bumps come from: the map of $\rho(\theta, \varphi)$ over the full angular range, and the six lobes of the equator.

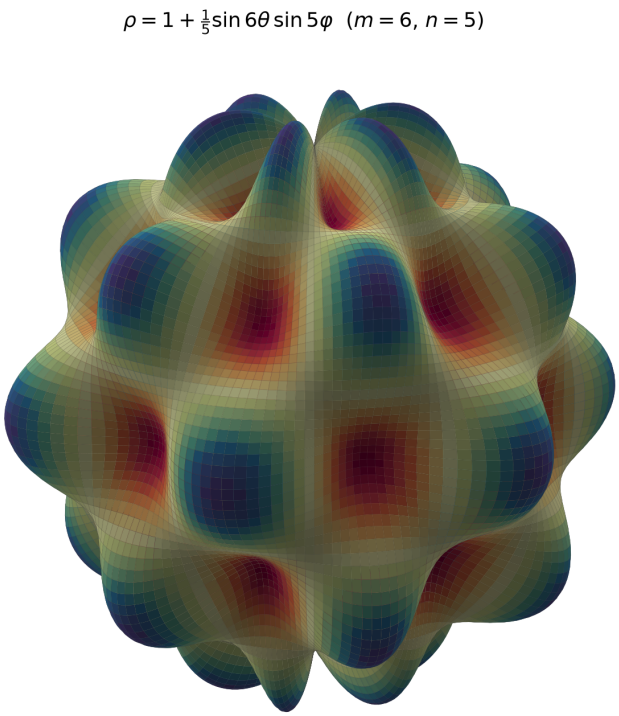


Figure 1: The bumpy sphere $\rho = 1 + \frac{1}{5}\sin 6\theta \sin 5\varphi$. The surface is coloured by ρ, from 0.8 (dark blue) to 1.2 (dark red). Six bumps lie around each parallel and five rows of bumps run from pole to pole.

How to solve problems of this kind

1. Write the surface as $\rho = \rho_{\max}(\theta, \varphi)$ and check that $\rho_{\max} > 0$. If it is, each ray from the origin meets the surface once, the solid is star shaped, and $0 \le \rho \le \rho_{\max}$ is a legal description of it.
2. Integrate in ρ first. The result is always $\frac{1}{3}\iint \rho_{\max}^3 \sin\varphi \, d\varphi \, d\theta$, so the problem becomes a double integral over a rectangle in the (θ, φ) plane.
3. Expand powers of $1 + u$ with the binomial theorem, then use $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ for the even powers. Every term separates into a function of θ times a function of φ.
4. Use the full angular ranges $0 \le \theta \le 2\pi$ and $0 \le \varphi \le \pi$. On a full range, the integral of an odd power of a sine or cosine over a whole number of periods is zero. Say which symmetry kills

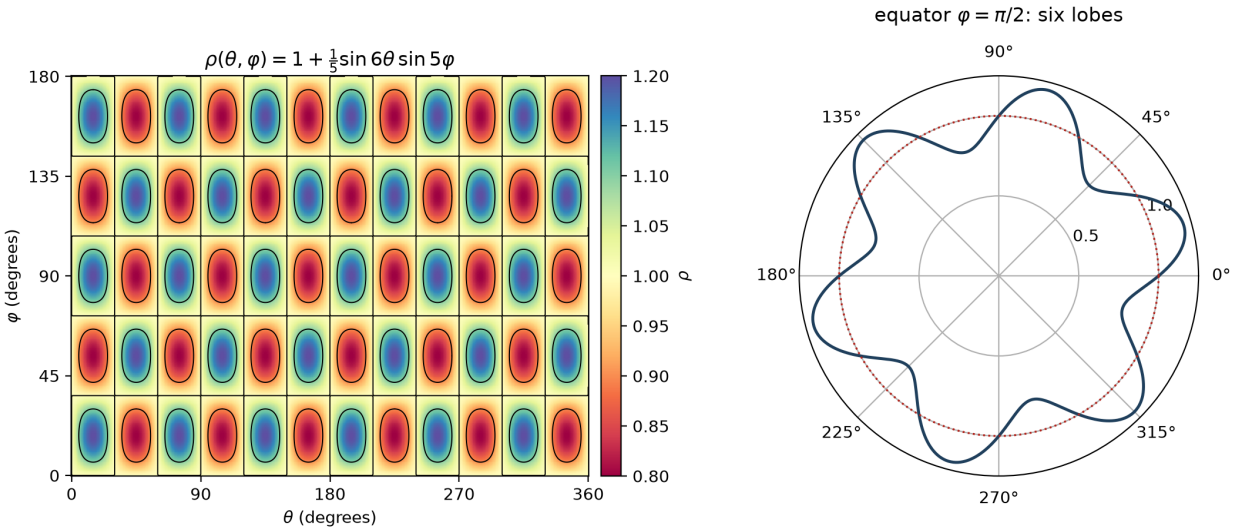


Figure 2: Left: the radius $\rho(\theta, \varphi)$ over the full range $0 \le \theta \le 2\pi$, $0 \le \varphi \le \pi$. The $\sin 6\theta$ factor gives six periods across a full turn (twelve half periods of alternating sign) and the $\sin 5\varphi$ factor gives five half periods from pole to pole; the black contours are $\rho = 0.9$, 1.0, 1.1. Right: the same function on the equator $\varphi = \pi/2$, drawn in polar form. The six lobes of the equator are the six bumps of Figure 1.

each term, instead of only writing that the term is zero.

5. Keep the exact value in the form $c\pi$ and then give a decimal. Compare it with the volume of the unit sphere, so the reader can judge the size.

6. Check with a computer algebra system twice, and with a numeric quadrature or a Monte Carlo estimate. Set a control case, such as $\rho_{\max} = 1$, to show that the code is right.